PATIENT CARE THEORY 2

UNIT 5, PART 1: Intro to Obstetrics

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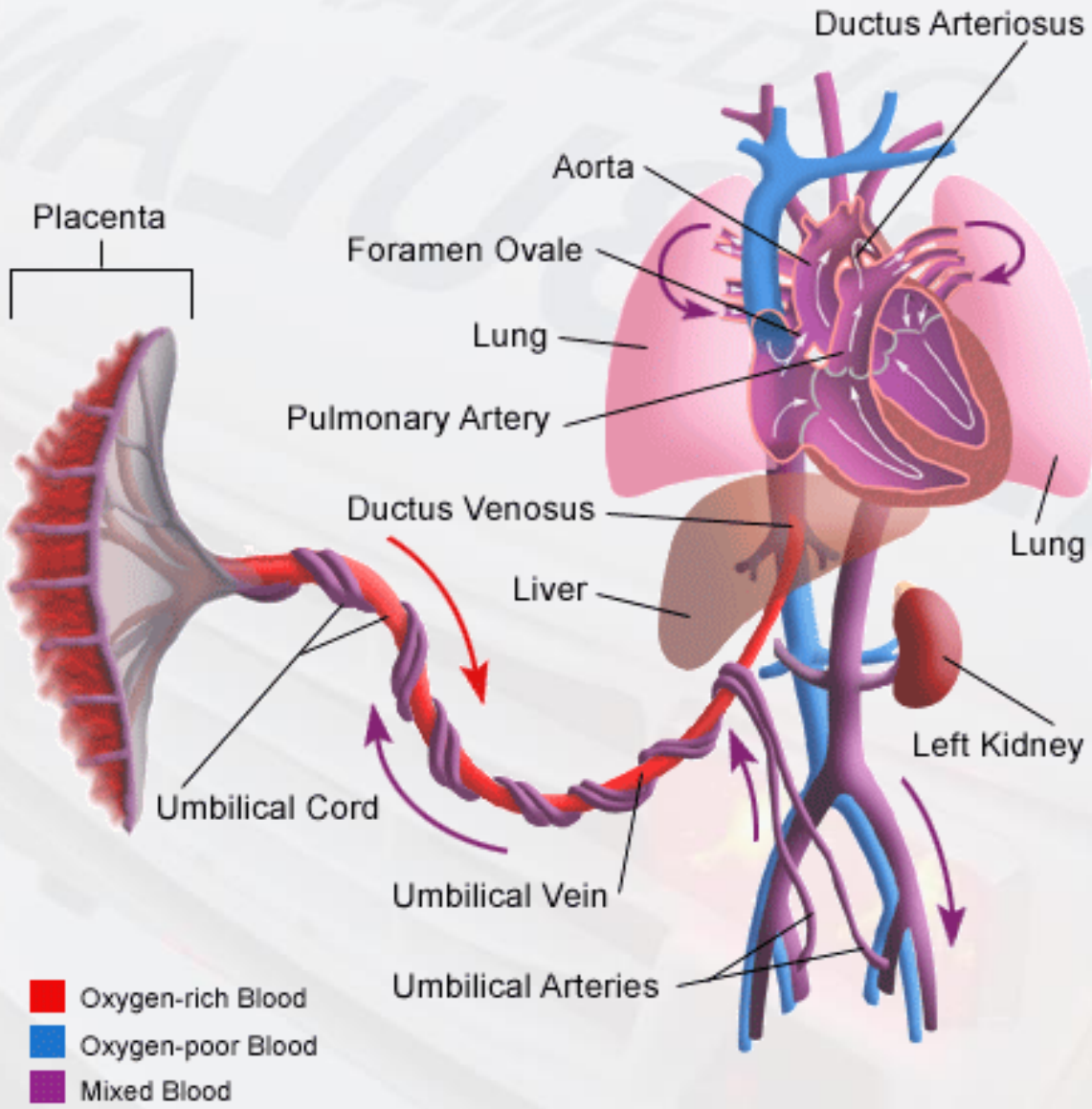
Placenta

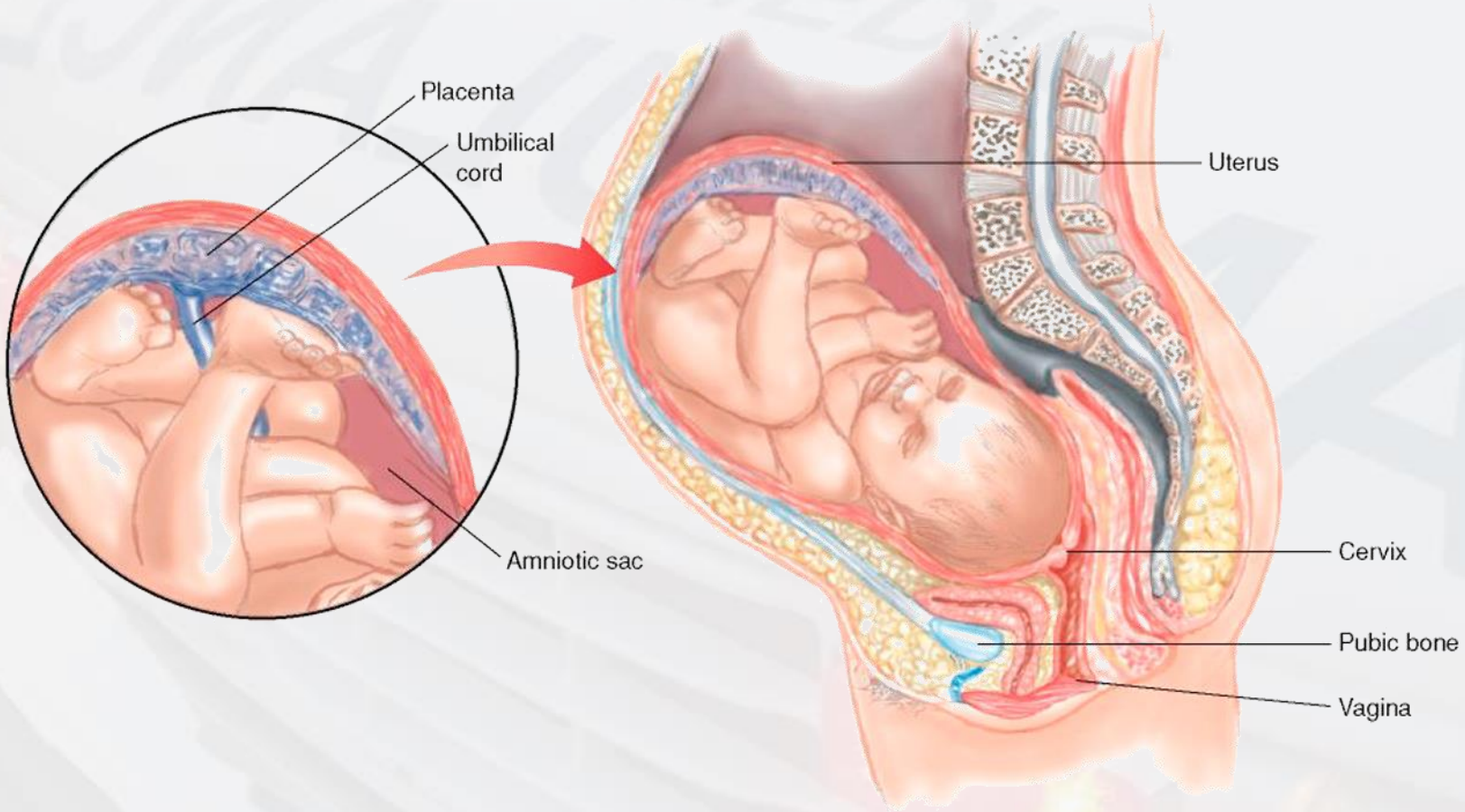
- develops three weeks after fertilization
- temporary blood rich organ
  - transfers heat
  - gas exchange
  - delivers nutrients, carries away wastes
  - endocrine gland – hormone production (HCG)
  - Protectant barrier – able to stop some harmful substances from crossing into the fetal circulation
- connected to fetus by umbilical cord
- expelled with afterbirth

Fetal Development

- Umbilical cord – connects the placenta to the fetus' vascular system
  - Vein transports oxygenated blood
    - **i.e. blood flow to the fetal heart from the mother**
  - Arteries return deoxygenated blood
    - i.e. away from the heart back to the mother
- Fetus develops within amniotic sac*
- Fluid surrounds and protects fetus
- Allows for fetal development

Fetal Circulation





"effacement" is thinning of the cervix

Physiologic Changes: Mother

- Reproductive system
  - Uterus increases in size
  - Vascular system
  - Formation of mucous plug in cervix
  - Estrogen causes vaginal mucosa to thicken
  - Breast enlargement

Physiologic Changes: Mother

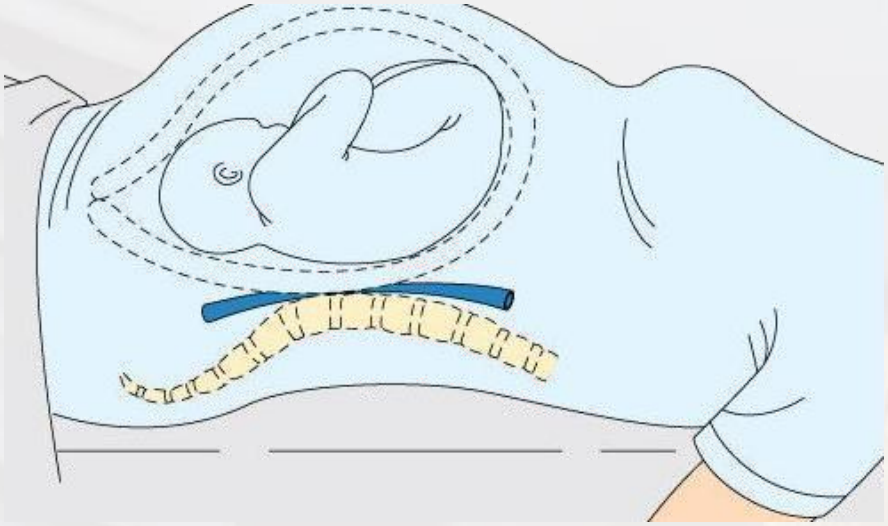
- Respiratory system
  - Progesterone causes a decrease in airway resistance.
    - Increase in oxygen consumption.
    - Increase in tidal volume.
    - Slight increase in respiratory rate
- Cardiovascular system
    - Cardiac output increases.
    - Blood volume increases by 40%-50%
      - facilitates maternal and fetal exchanges of respiratory gases, nutrients and metabolites
      - reduces the impact of maternal blood loss at delivery

Physiologic Changes: Mother

- Urinary system
  - Urinary frequency is common.
- Musculoskeletal system
  - Loosened pelvic joints.

Supine Hypotension

- Weight of uterus (3rd trimester) *may* compress the vena cava
- Decrease venous return → decreased CO
- Position of comfort on stretcher unless hypotension occurs → elevate right side 10-15°



"effacement" means?

A enlargement of the uterus

B thinning of the cervix

C increased heart beat during pregnancy

D development of a mucous plug

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What is the cause of supine hypotension?

A compression of the superior vena cava by the uterus

B a decrease in the size of the maternal heart

C compression of the inferior vena cava by the uterus

D decreased cardiac output due to changes in hormone levels

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C Compression of the inferior vena cava by the uterus

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Physiological Changes

Physiologic Changes: Fetus

- 2 weeks
  - Rapid cellular multiplication and differentiation

Physiologic Changes: Fetus

- 4 weeks
  - Fetal heart begins to beat

Physiologic Changes: Fetus

- 8 weeks
  - All body systems and external structures are formed
  - Size: ~ 3 cm (1.2”)

Physiologic Changes: Fetus

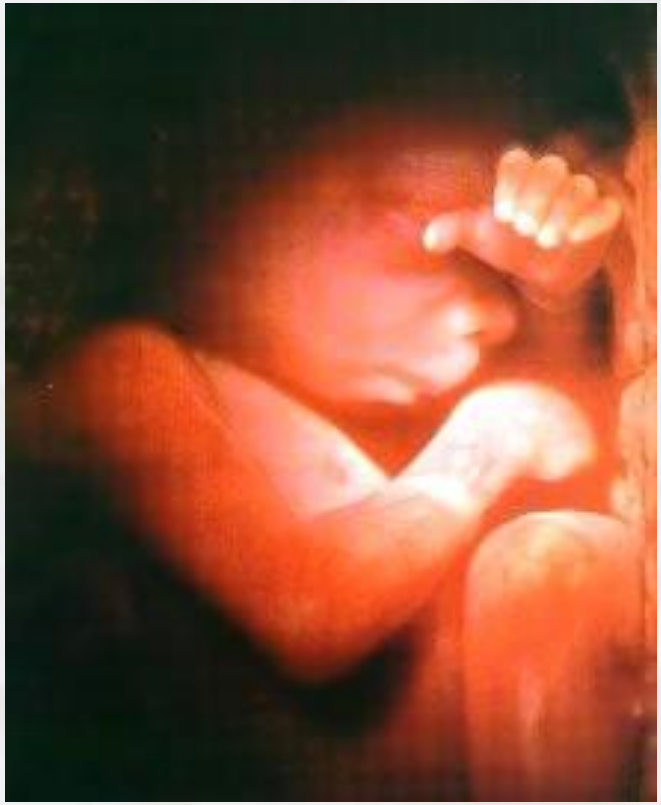
- 8-12 weeks
  - Fetal heart tones audible by doppler
  - Kidneys begin to produce urine
  - Size: 8 cm (3.2”)
  - Weight: 1.6 ounces
  - Fetus vulnerable to toxins

Physiologic Changes: Fetus

- 16 weeks
  - Sex can be determined
  - Swallowing amniotic fluid and producing meconium

Physiologic Changes: Fetus

- 20 weeks

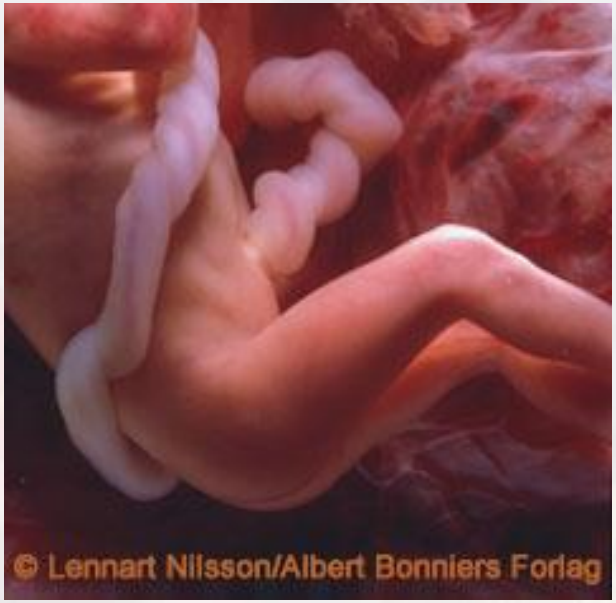


Physiologic Changes: Fetus

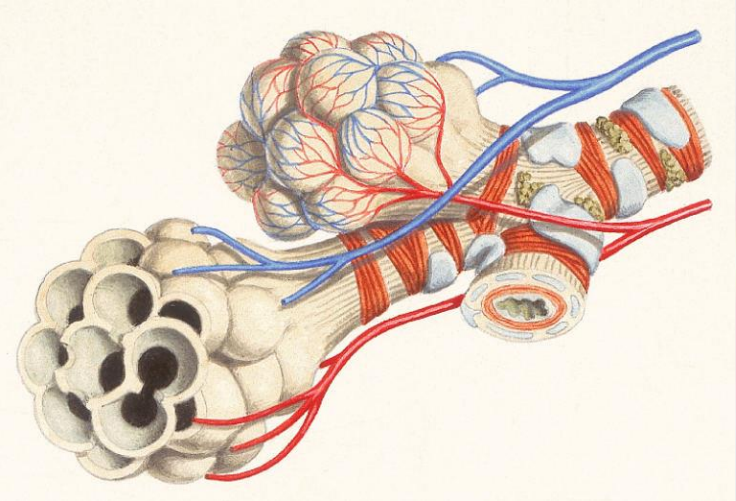
- 24 weeks
  - Increased activity
  - Begins respiratory movement (no air exchange)
  - Size: 28 cm (11.2”)
  - Weight: 1 lb, 10 ounces

Physiologic Changes: Fetus

- 28 weeks
  - Surfactant is formed
  - Eyes begin to open and close
  - Weight: 2-3 pounds



Alveolar surface tension



Physiologic Changes: Fetus

- 32 weeks
  - Bones are fully developed, but soft and flexible
  - Subcutaneous fat being deposited
  - Fingernails and toenails present

Physiologic Changes: Fetus

- 38-40 weeks
  - Full-term
  - Baby fills uterine cavity
  - Baby receives maternal antibodies

At what gestational age does the heart of the fetus begin to beat?

A 2 weeks

B 4 weeks

C 8 weeks

D 16 weeks

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At what gestational age does surfactant production begin?

A 16 weeks

B 20 weeks

C 28 weeks

D 32 weeks

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Describe how surfactant is like soap?

What gestational age is considered the minimum age of viability for survival?

A 16 weeks

B 24 weeks

C 28 weeks

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Management

- At 20 weeks gestation patients go straight to the L&D (Birthing Unit) – check with local policies
- Pregnant patients involved in minor trauma go to the hospital
- Position for comfort or left lateral tilt
- SpO2
- IV access prn
- O_2 if pregnancy complications or suspected complications with fetus
- +/- auscultate fetal heart sounds
- More information in next lesson…

QUESTIONS